

Two-Factor Arbitrage Pricing Theory Model

1 Project Overview

This document explains the mathematical structure behind the Excel workbook *Two-Factor Arbitrage Pricing Theory Model*. The workbook implements a two-factor version of Arbitrage Pricing Theory (APT) using monthly stock returns for two groups of firms: gold mining stocks and technology stocks. The two systematic factors are:

- the stock market excess return, used as the market factor;
- the excess return on the GLD ETF, used as a factor-mimicking portfolio for gold-price exposure.

The project has two main objectives:

1. estimate each stock's exposure to market risk and gold-price risk using an ordinary least squares regression;
2. use the estimated sector-average factor loadings and the APT pricing relation to compute expected monthly excess returns for the gold mining sector and the technology sector.

The key idea is that expected excess return is compensation for exposure to priced systematic risk factors. Idiosyncratic risk may affect an individual stock's realized return, but it is not supposed to command a systematic risk premium in the APT framework.

2 Arbitrage Pricing Theory Background

APT begins with a factor representation of returns. In a one-factor model, the random return of asset i may be written as

$$\tilde{r}_i = \bar{r}_i + b_i \tilde{f} + \tilde{\epsilon}_i, \quad (1)$$

where $\bar{r}_i = \mathbb{E}[\tilde{r}_i]$ is the expected return, $\tilde{f}$ is a common systematic shock, b_i is the factor loading, and $\tilde{\epsilon}_i$ is the idiosyncratic shock. The factor shock and the idiosyncratic shock are assumed to have zero mean:

$$\mathbb{E}[\tilde{f}] = 0, \quad \mathbb{E}[\tilde{\epsilon}_i] = 0. \quad (2)$$

If the idiosyncratic shocks are uncorrelated with the systematic factor and uncorrelated across assets, then the variance decomposes into a systematic and an idiosyncratic component:

$$\text{Var}(\tilde{r}_i) = b_i^2 \sigma_f^2 + \text{Var}(\tilde{\epsilon}_i). \quad (3)$$

The covariance between two assets is driven by their common exposure to the systematic factor:

$$\text{Cov}(\tilde{r}_i, \tilde{r}_j) = b_i b_j \sigma_f^2. \quad (4)$$

A multifactor model generalizes this idea to K systematic factors:

$$\tilde{r}_i = \bar{r}_i + \sum_{k=1}^K b_{i,k} \tilde{f}_k + \tilde{\epsilon}_i. \quad (5)$$

Here $b_{i,k}$ measures the sensitivity of asset i to factor k .

3 Portfolios, Diversification, and Systematic Risk

For a portfolio p with weights w_i , the random return is

$$\tilde{r}_p = \sum_i w_i \tilde{r}_i. \quad (6)$$

Substituting the multifactor return model gives

$$\tilde{r}_p = \bar{r}_p + \sum_{k=1}^K b_{p,k} \tilde{f}_k + \tilde{\epsilon}_p, \quad (7)$$

where

$$\bar{r}_p = \sum_i w_i \bar{r}_i, \quad b_{p,k} = \sum_i w_i b_{i,k}, \quad \tilde{\epsilon}_p = \sum_i w_i \tilde{\epsilon}_i. \quad (8)$$

If the idiosyncratic shocks are uncorrelated across assets, then the idiosyncratic variance of the portfolio is

$$\text{Var}(\tilde{\epsilon}_p) = \sum_i w_i^2 \text{Var}(\tilde{\epsilon}_i). \quad (9)$$

For an equally weighted portfolio with N assets, $w_i = 1/N$, this becomes

$$\text{Var}(\tilde{\epsilon}_p) = \frac{1}{N^2} \sum_{i=1}^N \text{Var}(\tilde{\epsilon}_i). \quad (10)$$

Thus, as the number of assets grows and the weights remain well spread out, the idiosyncratic component becomes negligible:

$$\tilde{\epsilon}_p \approx 0. \quad (11)$$

A well-diversified portfolio is therefore exposed mainly to systematic factors:

$$\tilde{r}_p \approx \bar{r}_p + \sum_{k=1}^K b_{p,k} \tilde{f}_k. \quad (12)$$

This is the central economic logic behind APT: diversification can eliminate firm-specific shocks, but it cannot eliminate systematic factor exposures.

4 The APT Pricing Relation

In a frictionless market, well-diversified portfolios with the same factor loadings must have the same expected excess return. Otherwise, an investor could form a zero-factor-exposure portfolio with positive payoff, creating an arbitrage opportunity.

For a single-factor model, the APT relation is

$$\bar{r}_p - r_f = \lambda b_p, \quad (13)$$

where λ is the factor risk premium, or the price of one unit of factor risk. In a multifactor setting,

$$\bar{r}_p - r_f = \lambda_1 b_{p,1} + \lambda_2 b_{p,2} + \cdots + \lambda_K b_{p,K}. \quad (14)$$

Equivalently,

$$\bar{r}_p - r_f = \sum_{k=1}^K \lambda_k b_{p,k}. \quad (15)$$

The phrase “price of risk times quantity of risk” summarizes the interpretation:

$$\text{risk premium} = \sum_k (\text{factor risk price}) \times (\text{factor exposure}). \quad (16)$$

5 The Two-Factor Model Used in the Workbook

The workbook applies APT with two systematic factors. For stock i in month t , define excess returns as

$$R_{i,t} = r_{i,t} - r_{f,t}, \quad (17)$$

$$M_t = r_{\text{MKT},t} - r_{f,t}, \quad (18)$$

$$G_t = r_{\text{GLD},t} - r_{f,t}. \quad (19)$$

The empirical regression model is

$$R_{i,t} = \alpha_i + \beta_{i,\text{MKT}} M_t + \beta_{i,\text{GLD}} G_t + \epsilon_{i,t}. \quad (20)$$

The coefficients have the following meaning:

- α_i is the intercept, often interpreted as the part of average excess return not explained by the two factors;
- $\beta_{i,\text{MKT}}$ measures exposure to stock market excess returns;
- $\beta_{i,\text{GLD}}$ measures exposure to gold-price-related excess returns;
- $\epsilon_{i,t}$ is the residual return not explained by the two-factor model.

A positive $\beta_{i,\text{GLD}}$ means that the stock tends to move with the gold factor. A $\beta_{i,\text{MKT}}$ close to 1 means that the stock tends to move approximately one-for-one with the market factor.

6 OLS Estimation

For each stock, the workbook estimates Equation (20) using ordinary least squares. Let

$$y_i = \begin{bmatrix} R_{i,1} \\ R_{i,2} \\ \vdots \\ R_{i,T} \end{bmatrix} \quad (21)$$

be the vector of stock excess returns, and let

$$X = \begin{bmatrix} 1 & M_1 & G_1 \\ 1 & M_2 & G_2 \\ \vdots & \vdots & \vdots \\ 1 & M_T & G_T \end{bmatrix}. \quad (22)$$

Then the coefficient vector is

$$\theta_i = \begin{bmatrix} \alpha_i \\ \beta_{i,\text{MKT}} \\ \beta_{i,\text{GLD}} \end{bmatrix}. \quad (23)$$

The OLS estimator is

$$\hat{\theta}_i = (X^\top X)^{-1} X^\top y_i. \quad (24)$$

In the Excel workbook, this regression is implemented with Excel's `LINEST` function. Conceptually, the function estimates the intercept and the two slope coefficients for each stock.

7 Estimated Stock-Level Coefficients

The estimated coefficients are summarized below. The gold mining stocks have strong positive exposure to the GLD factor, while the technology stocks have market betas close to one and slightly negative exposure to the GLD factor.

Stock	β_{GLD}	β_{MKT}	α
Kinross Gold Corporation	1.920	0.300	−0.88%
AngloGold Ashanti	1.824	0.095	−0.73%
Barrick Gold Corporation	1.811	0.161	−0.82%
Newmont Corporation	1.449	0.202	−0.64%
Apple Inc.	−0.072	1.079	2.10%
Microsoft Corporation	−0.229	0.923	0.88%
Oracle Corporation	−0.075	0.979	0.36%
Intel Corporation	−0.297	0.946	0.53%

The interpretation is straightforward:

- Gold mining stocks are highly sensitive to the gold factor because their cash flows and valuations are naturally related to gold prices.
- Technology stocks are mainly exposed to broad market movements and have small, negative estimated exposure to the gold factor.

8 Sector-Level Average Factor Loadings

The workbook then averages the estimated coefficients within each sector:

$$\bar{\beta}_{s,k} = \frac{1}{n_s} \sum_{i \in s} \hat{\beta}_{i,k}, \quad (25)$$

where s denotes the sector and n_s is the number of stocks in that sector.

Sector	$\bar{\beta}_{\text{GLD}}$	$\bar{\beta}_{\text{MKT}}$	$\bar{\alpha}$
Gold Mining Stocks	1.751	0.189	−0.8%
Technology Stocks	−0.168	0.982	1.0%

The sector averages show a clear contrast. Gold mining stocks have strong gold exposure and weak market exposure. Technology stocks have strong market exposure and weak negative gold exposure.

9 Forecasting Expected Excess Returns

In the forecasting step, the workbook applies the exact two-factor APT relation. Let the expected monthly excess returns on the factors be

$$\lambda_{\text{MKT}} = 0.75\%, \quad \lambda_{\text{GLD}} = 0.15\%. \quad (26)$$

Under the exact two-factor APT model, the expected excess return of sector s is

$$\mathbb{E}[R_s] = \lambda_{\text{MKT}} \bar{\beta}_{s,\text{MKT}} + \lambda_{\text{GLD}} \bar{\beta}_{s,\text{GLD}}. \quad (27)$$

Notice that the historical regression intercept α is not used in this final APT forecast. In the exact APT pricing relation, expected excess returns are determined by priced systematic factor exposures.

9.1 Gold Mining Sector

Using the sector-average factor loadings,

$$\bar{\beta}_{\text{MKT}} = 0.189, \quad \bar{\beta}_{\text{GLD}} = 1.751. \quad (28)$$

Therefore,

$$\mathbb{E}[R_{\text{Gold}}] = (0.75\%)(0.189) + (0.15\%)(1.751) \quad (29)$$

$$= 0.14175\% + 0.26265\% \quad (30)$$

$$= 0.40440\% \quad (31)$$

$$\approx 0.40\% \text{ per month.} \quad (32)$$

9.2 Technology Sector

Using the sector-average factor loadings,

$$\bar{\beta}_{\text{MKT}} = 0.982, \quad \bar{\beta}_{\text{GLD}} = -0.168. \quad (33)$$

Therefore,

$$\mathbb{E}[R_{\text{Tech}}] = (0.75\%)(0.982) + (0.15\%)(-0.168) \quad (34)$$

$$= 0.73650\% - 0.02520\% \quad (35)$$

$$= 0.71130\% \quad (36)$$

$$\approx 0.71\% \text{ per month.} \quad (37)$$

Sector	Expected Monthly Excess Return
Gold Mining Stocks	0.40%
Technology Stocks	0.71%

10 Economic Interpretation

The estimated model says that the two sectors are exposed to different systematic risks.

The gold mining sector has a large positive GLD beta. Its expected excess return is therefore strongly affected by the expected premium on the gold factor. However, its market beta is low, so the market factor contributes less to its expected excess return.

The technology sector has a market beta close to one. Its expected excess return is therefore mostly driven by the expected market premium. The negative GLD beta slightly reduces the technology sector's forecasted expected excess return because the GLD factor premium is positive while the sector's gold exposure is negative.

This can be seen by decomposing the forecasts:

Sector	Market Contribution	GLD Contribution	Total
Gold Mining Stocks	0.14%	0.26%	0.40%
Technology Stocks	0.74%	-0.03%	0.71%

11 Conclusion

The workbook is an empirical implementation of a two-factor APT model. It first estimates stock-level factor loadings with OLS regressions and then averages those loadings by sector. Finally, it applies the APT pricing relation to transform factor exposures into expected monthly excess returns.

The main results are:

- Gold mining stocks have high positive exposure to the GLD factor and low exposure to the market factor.
- Technology stocks have market betas close to one and slightly negative exposure to the GLD factor.
- With factor risk premia of 0.75% for the market and 0.15% for GLD, the expected monthly excess return is approximately 0.40% for the gold mining sector and 0.71% for the technology sector.

Mathematically, the project is built around the APT equation

$$\bar{r}_p - r_f = \sum_{k=1}^K \lambda_k b_{p,k} \quad (38)$$

and its two-factor empirical version

$$\mathbb{E}[R_s] = \lambda_{\text{MKT}} \bar{\beta}_{s,\text{MKT}} + \lambda_{\text{GLD}} \bar{\beta}_{s,\text{GLD}}. \quad (39)$$